Department of Biomedical Engineering

המחלקה להנדסה ביו-רפואית

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הפקולטה להנדסה ע"ש איבי ואלדר פליישמן אוניברסיטת תל-אביב

The effect of different concentrations of Gelatin substrate compositions on cell migration

Presented by
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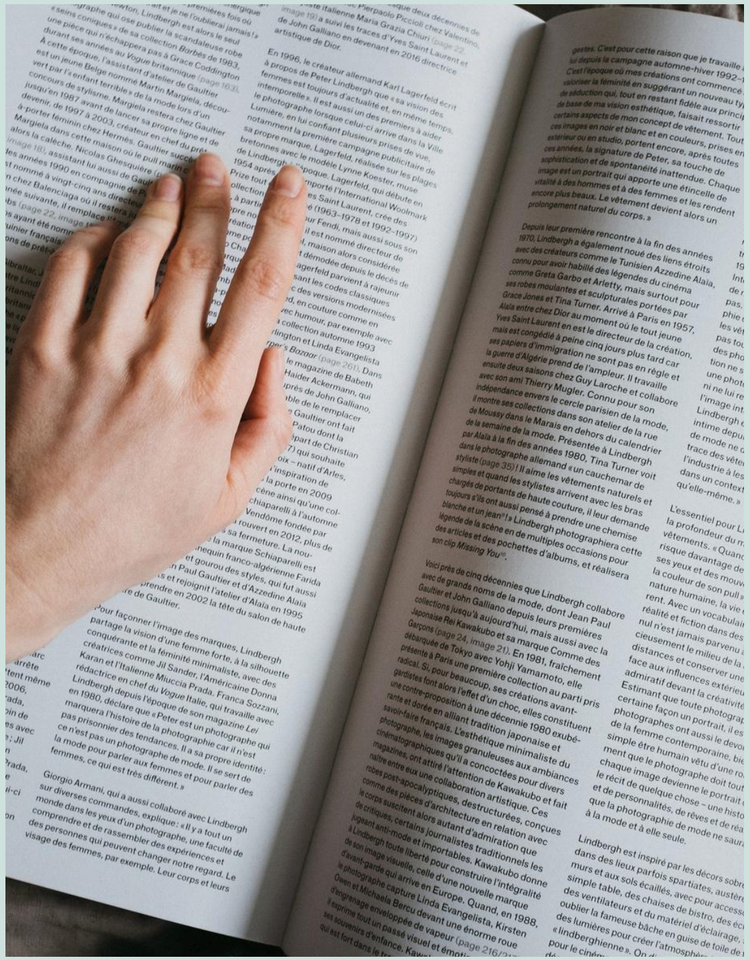
13 June 2023

Research Methods in Cell and Tissue Engineering



Literature Review

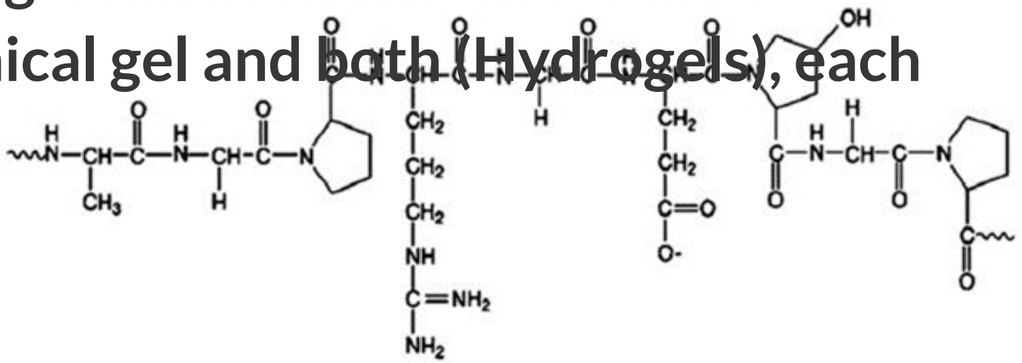
Substrate of cell culture

- Substrate of cell culture is the surface on which the cells grow and are attached to.

- The properties of cell culture substrates that affect cell behavior include growth, differentiation, spreading, and migration.

Gelatin

- Gelatin is a type of substrate used for cell culture in vitro
- It is a protein produced by hydrolyzing animal collagen, usually from connective tissues and bones.
- At around 30-40◦C, the gelatin transfers from a gel state to a solution and since the optimal temperature for cell culture growth is 37◦C, this feature of the gelatin must be taken in consideration.
- There are different types of gelatin networks that can be achieved: physical gel, chemical gel and both (Hydrogels), each has its own properties.



Cell migration

- Cell migration is the movement of a single cell or a group of cells in response to intracellular signaling cascades which can be received chemically or mechanically.

- The way in which cells migrate is largely influenced by the type of cell and the environment that guides their movement.

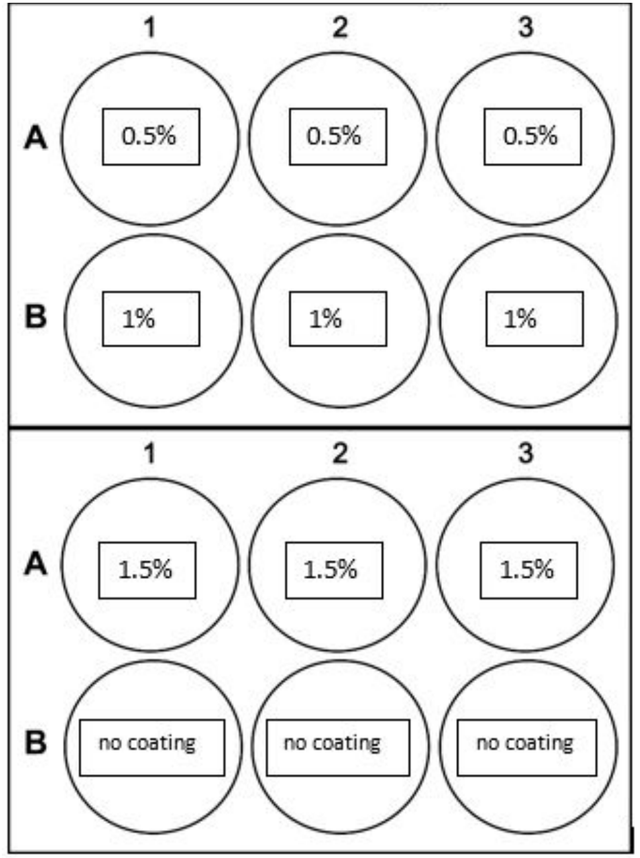
- While working in vitro, the environment of the cells is affected by the selected substrate composition.

Methods

Methods

Workflow Overview

- **Materials Preparation:**
  - Growth Medium
  - Gelatin Solutions (0%, 0.5%, 1%, 1.5%)
  - Cutting PDMS into structures
- **Thawing cells**
- **Passaging cells (sub-culturing)**
- **Cell Counting**
- **Wells Covering with Gelatin solutions**
- **Wells Covering with PDMS**
- **Cell Seeding**
- **Cells Imaging**



Computational Methods

- **The cell density in each well was calculated following the equation:**

$$Density = \frac{number\ of\ \text{cells}}{area}\ [\frac{cells}{\mu m^2}]$$

- **The ratio between the inside and the outside average densities was measured using the equation:**

$$Ratio = \frac{Averaged_Density_{\ out}}{Averaged_Density_{\ in}}$$

- **The difference between the ratios of the different time points was calculated using the equation:**

$$\text{D}ifference = \frac{Ratio_i - Ratio_j}{time\ difference}$$

Computational Methods

- **The general migration rate for C2C12 cells was assessed using the following equation (without reference to the different gelatin concentrations):**

$$\text{Migration rate} = \frac{total\ distance}{time}$$

Differences Between the 1st & 2nd Sessions

Gelatin Solutions

1st session:

0%, 0.5%, 1%, 1.5%

2nd session:

0%, 0.5%, 1%

PDMS structures

1st session:

circles

2nd session:

rectangles (1.5x3 cm size)

+ Cell Type: from **Adipocytes** to **C2C12**

Wells Covering

1st session:

- no pre-warming gelatin

2nd session:

- pre-warming gelatin

Cells Imaging

1st session:

- two sides of the outline
- no signing of locations

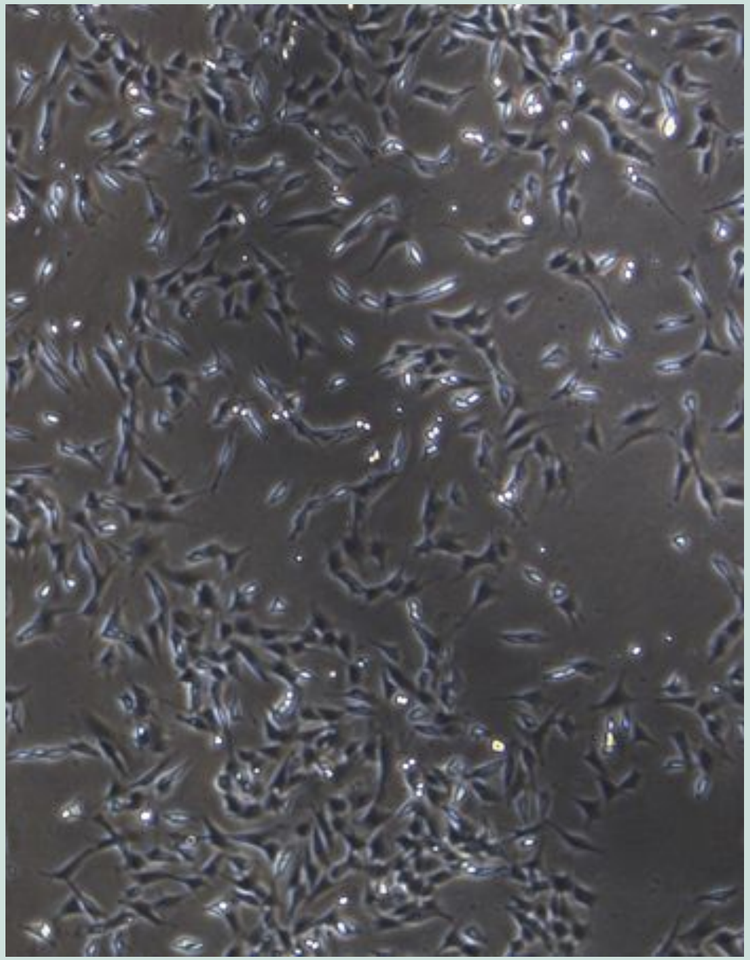
2nd session:

- inner and outer areas
- signing imaging locations

Computational Methods

ImageJ image processing:

MATLAB code for cell counting:



Results and Discussion

Comparison between 1st & 2nd sessions' results

Microscope Enlargement

4X instead of 10X

Methods Improvements

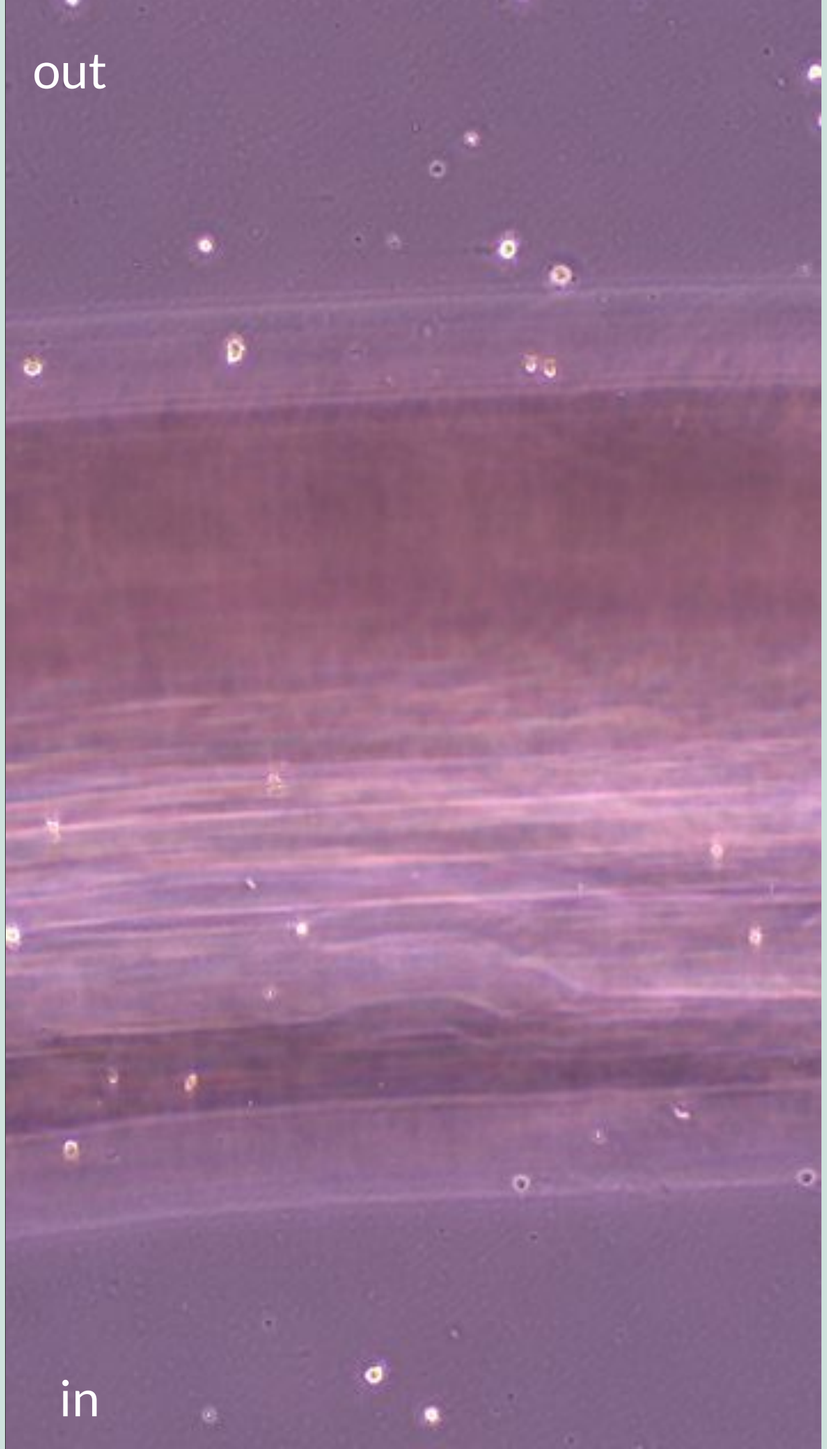
As mentioned in the Methods section

Cell Type

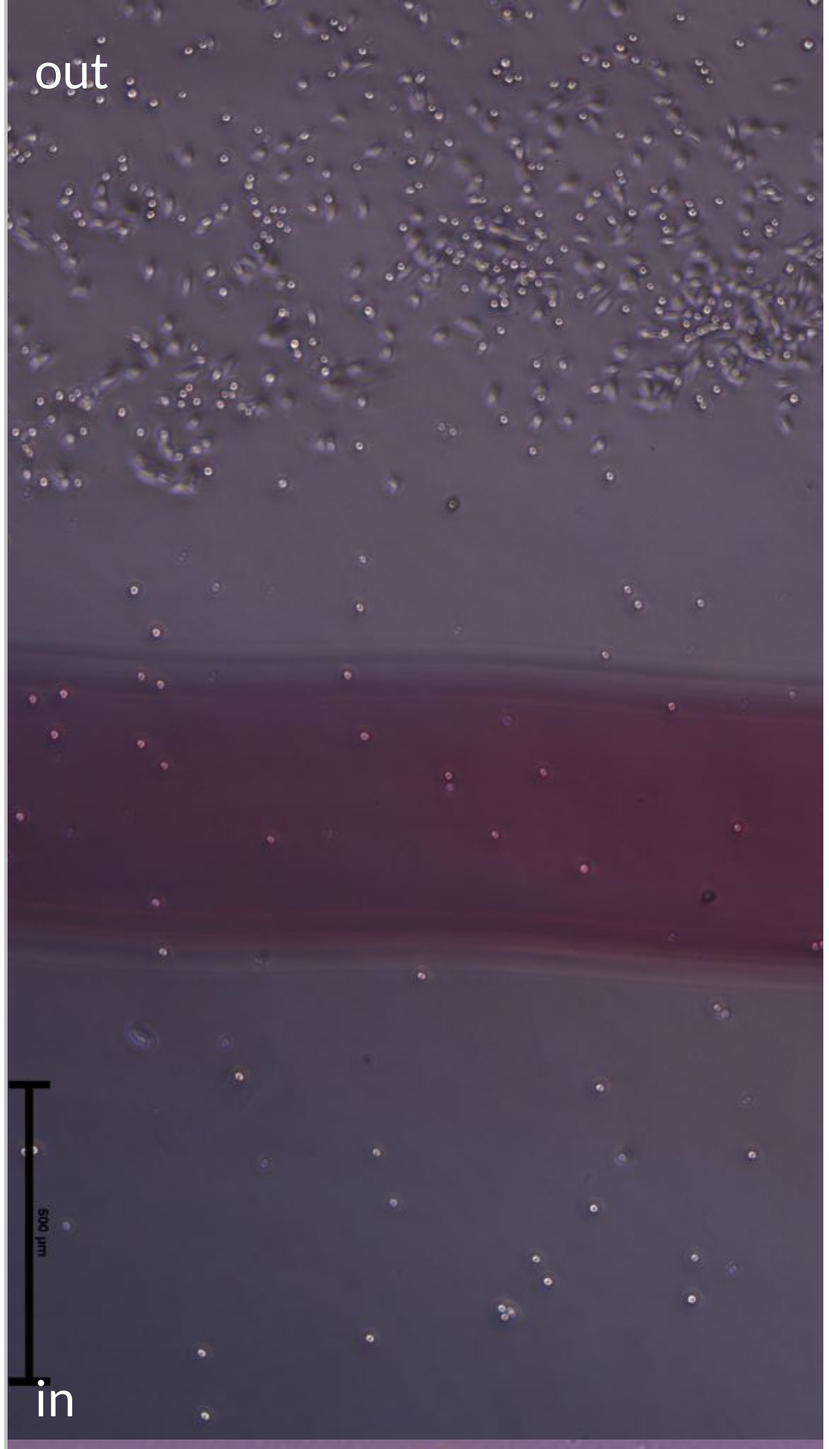
C2C12 instead of Adipocytes

day 0

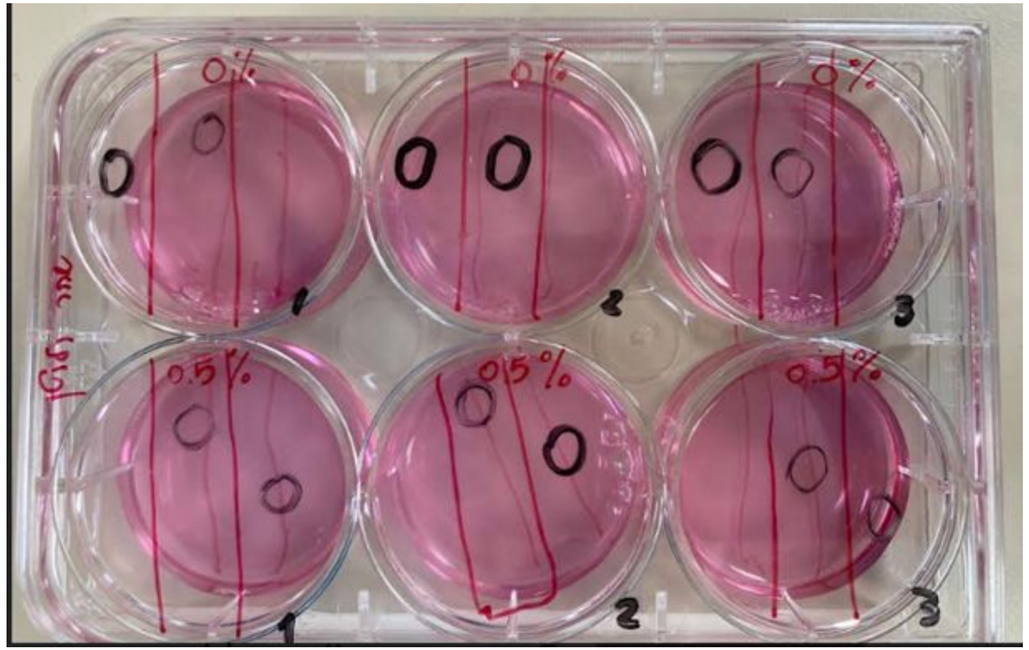
1st Experiment



2nd Experiment



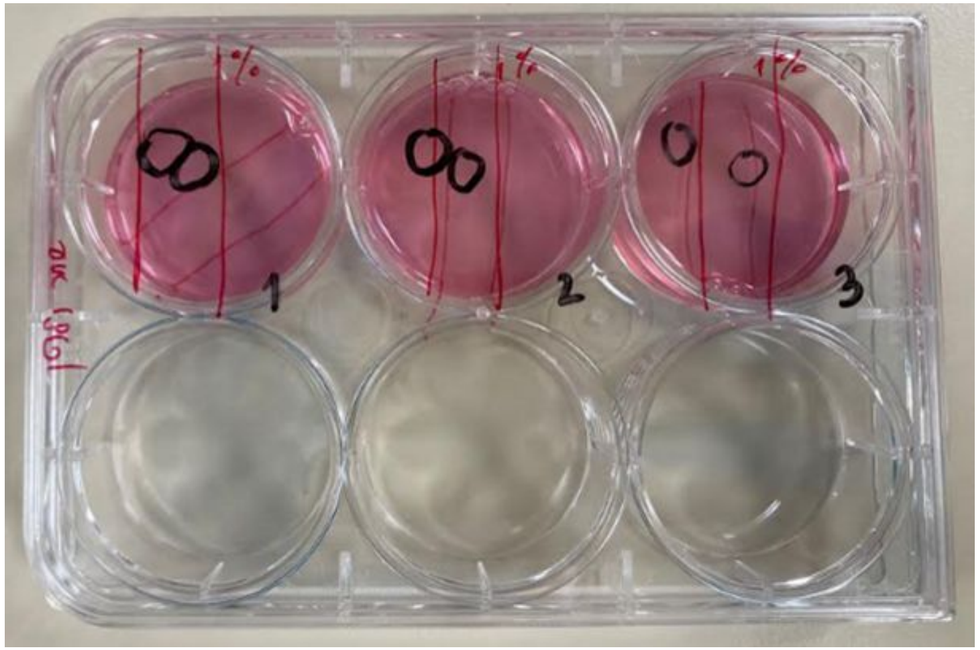
The marked wells after PDMS removal



0% Gelatin Solution

&

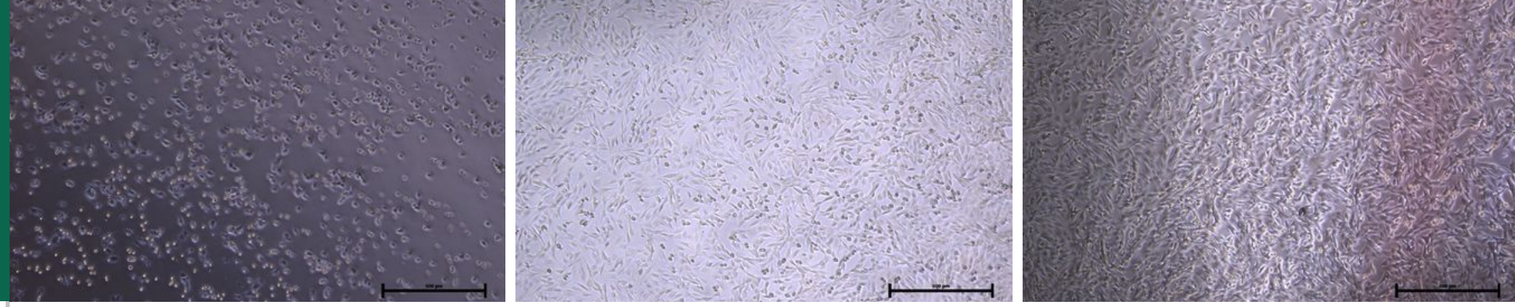
0.5% Gelatin Solution



1% Gelatin Solution

Most of the PDMS didn’t move, except for one in the 1% Gelatin concentration

0.5% Gelatin outer area over time:



Visually comparing cells' density over time

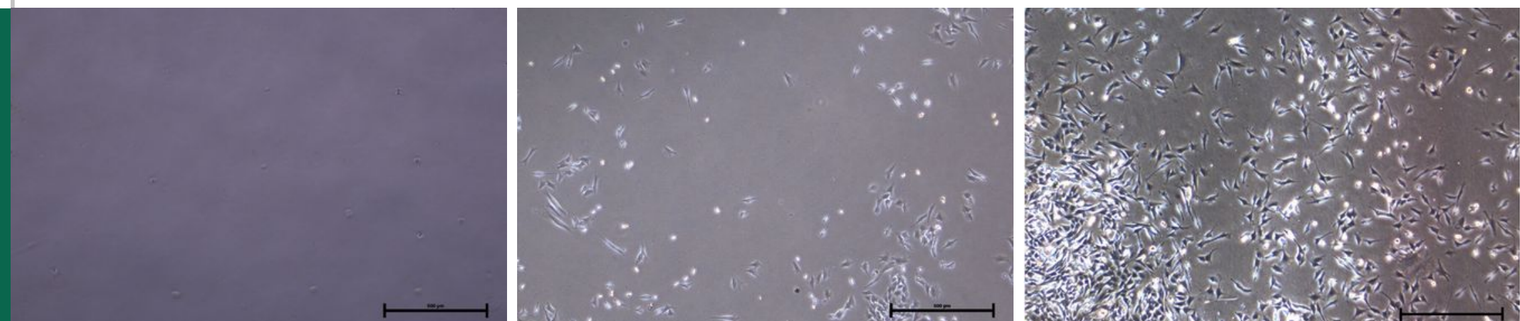
0.5% Gelatin Solution

1. Both outer and inner area's densities increased over time

2. The cells' morphology change over time

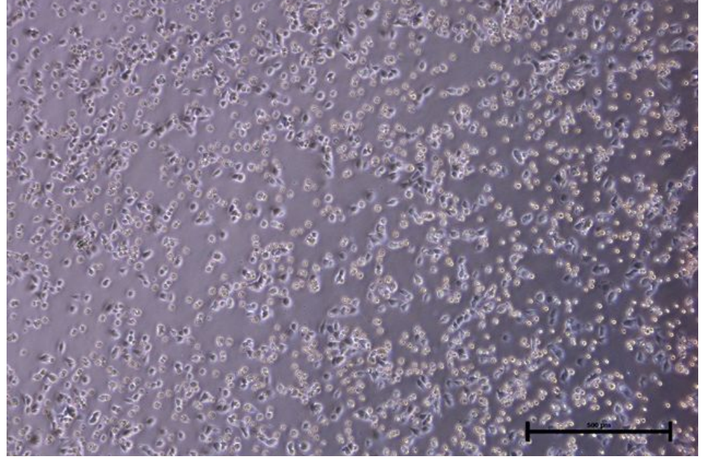
3. The densities of the outer area are larger than the densities of the inner area

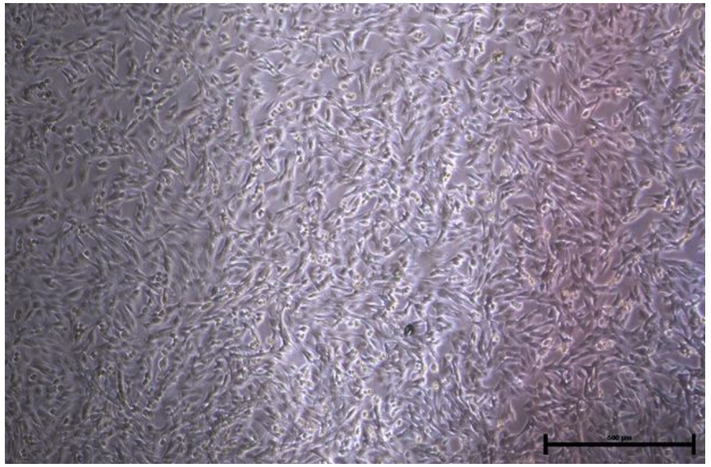
0.5% Gelatin inner area over time:

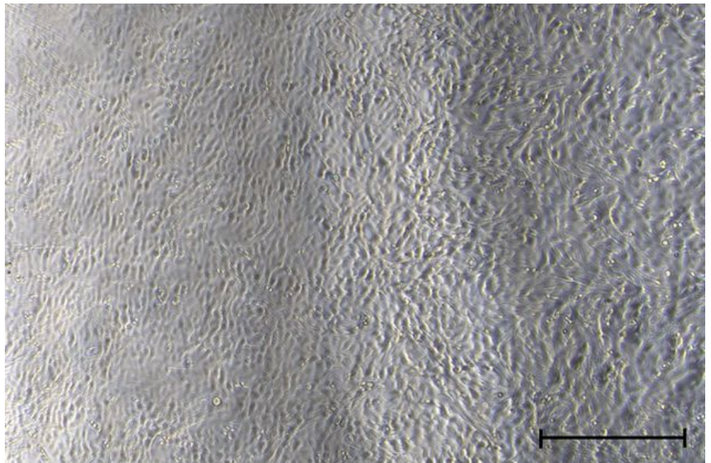


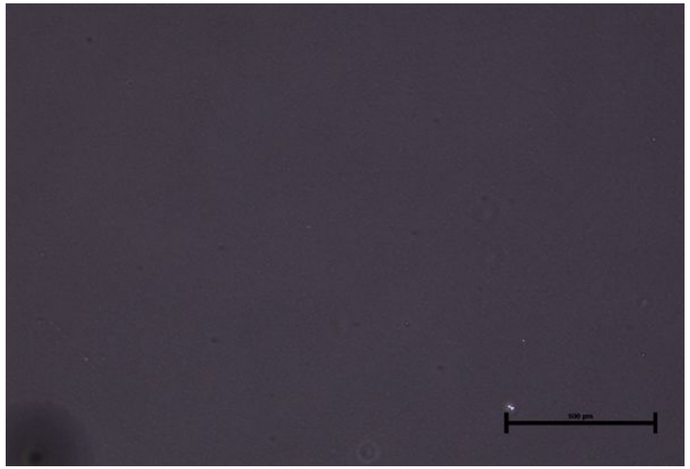
1% Gelatin Solution

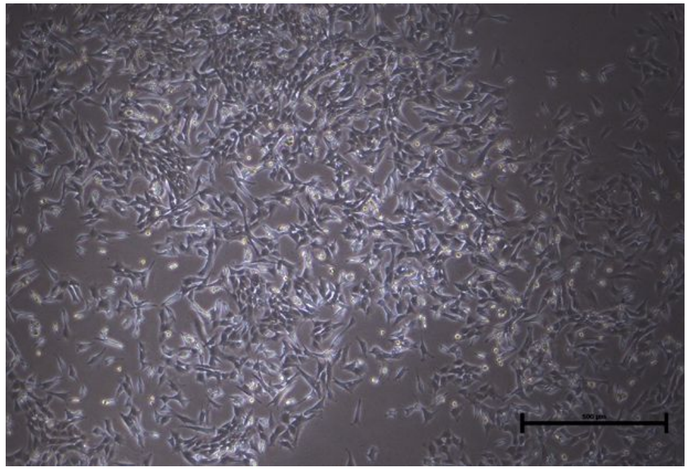
Outer area over time

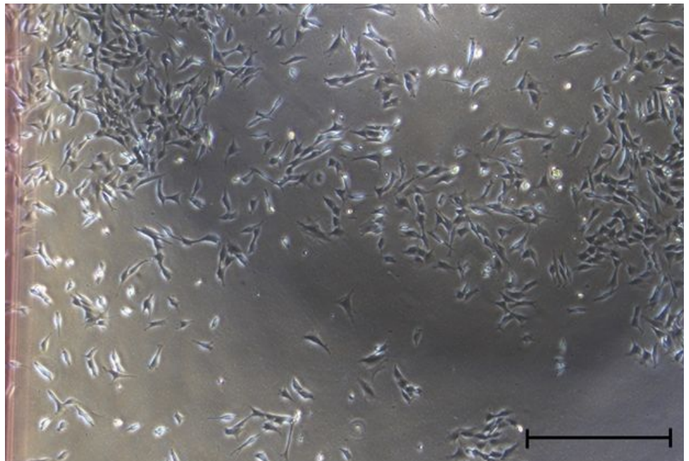












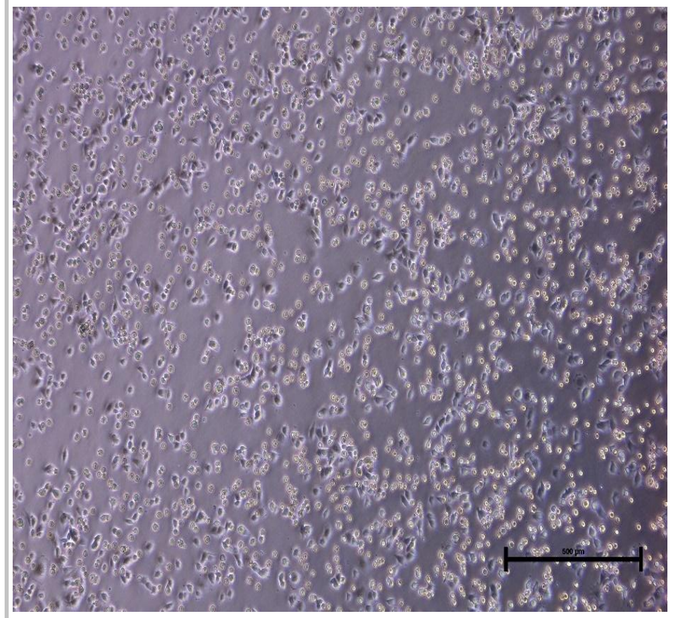
1% Gelatin Solution

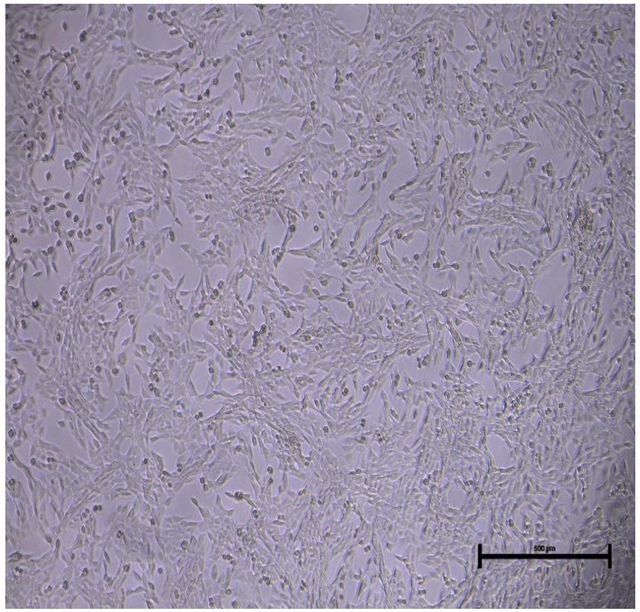
Inner area over time

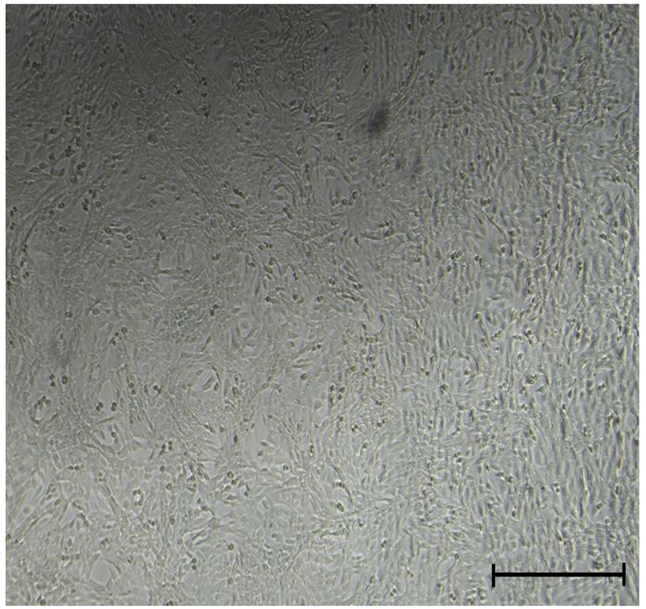
Outer area

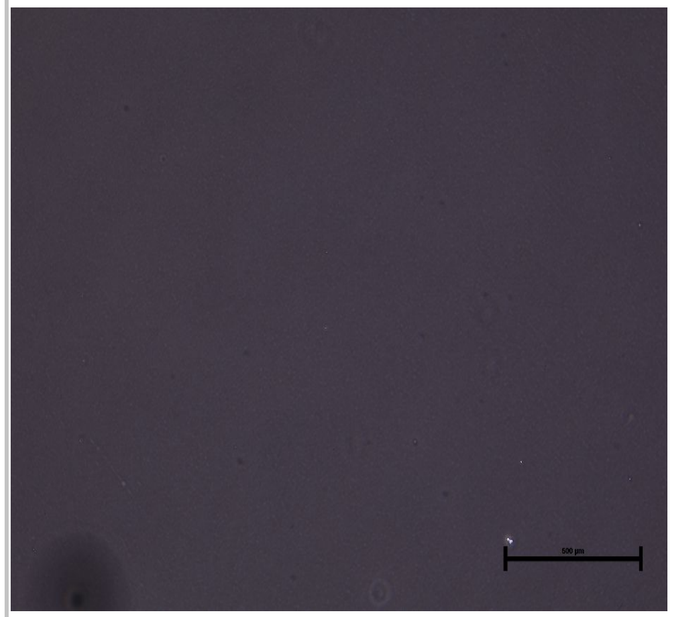
Control Group: 0% Gelatin

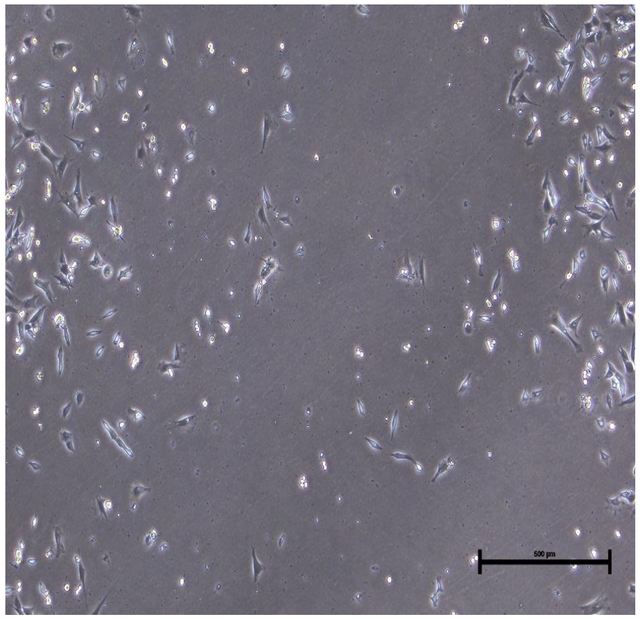
Inner area

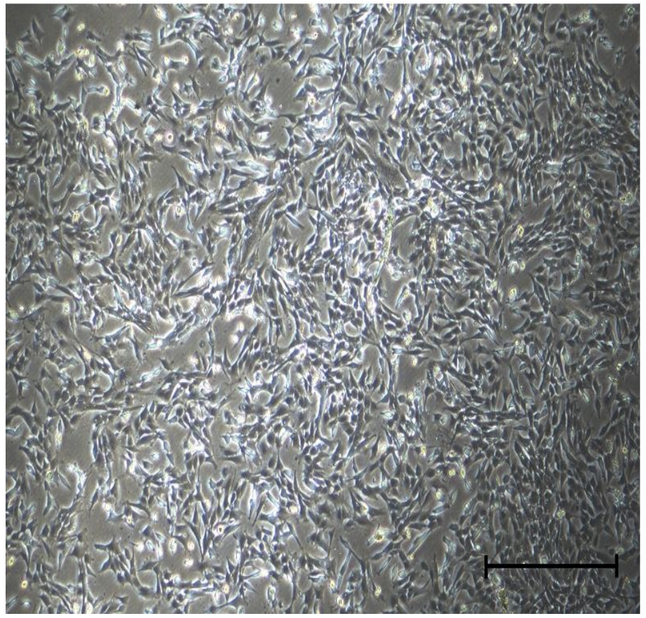












2nd Experiment

Well	Ratio, day 0 (Average out /Average in)	Ratio, day 1, morning (Average out /Average in)	Difference [Difference/ hour]
0% Gelatin	28.166	3.368	1.653
0.5% Gelatin	28.255	2.619	1.709
1% Gelati	56.11	1.225	3.657

Calculated differences between

day 0 -> day 1 morning

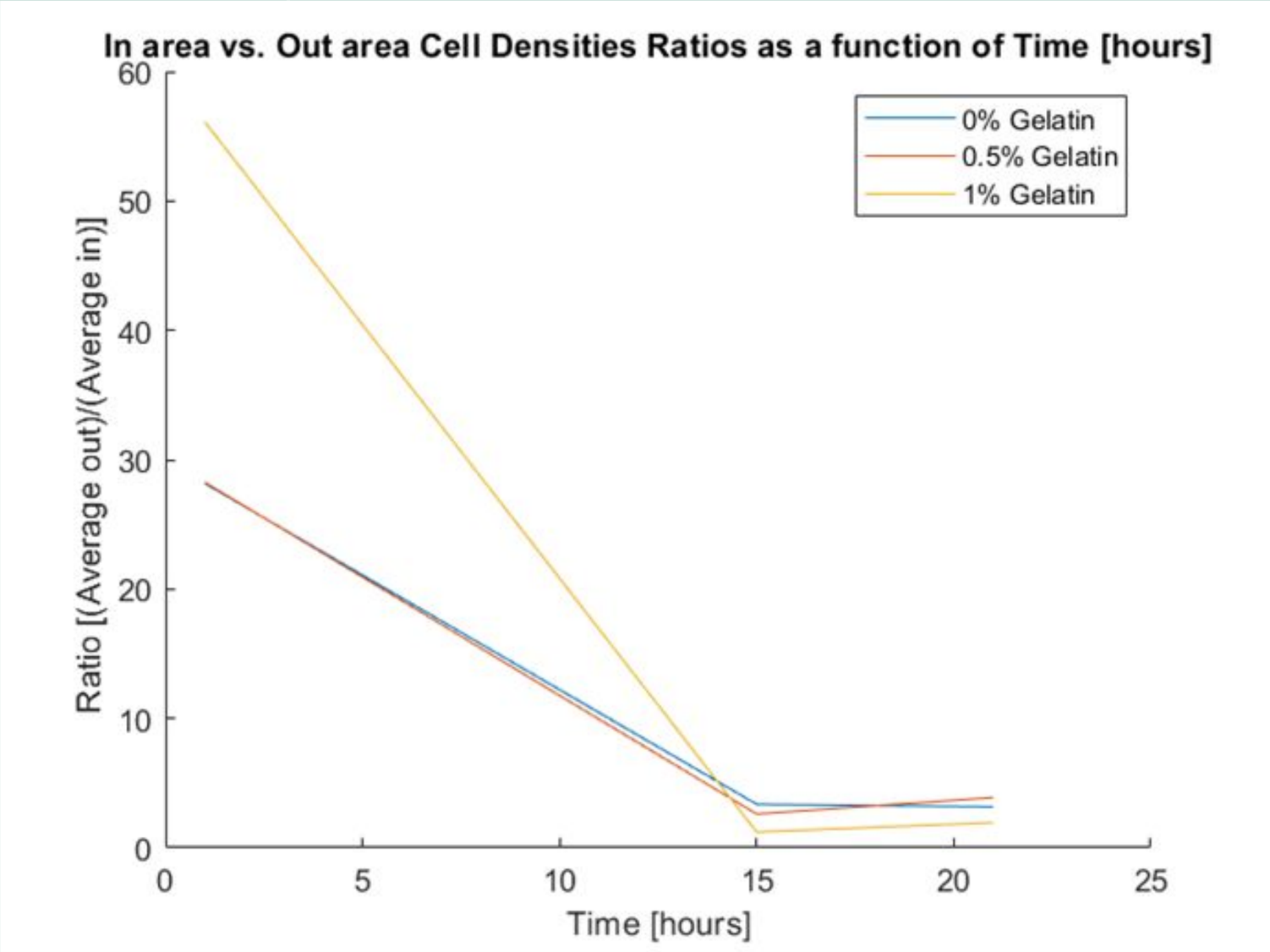
Well	Ratio, day 1, morning (Average out /Average in)	Ratio, day 1, noon (Average out /Average in)	Difference [Difference/ hour]
0% Gelatin	3.368	3.169	0.033
0.5% Gelatin	2.619	3.884	-0.211
1% Gelatin	1.225	1.954	-0.122

Calculated differences between

day 1 morning -> day 1 noon

Densities Ratios over time

relations between the different densities' ratios over time for the different gelatin concentrations



Cell densities ratios as a function of time [hours] for the different gelatin concentrations [0%, 0.5%, 1%]

Conclusions

Main Findings

A high gelatin concentration contributes to C2C12 adhesion

Cell migration was favored by the highest concentration of gelatin (1%)

Most of the cell's migration occurred during the initial 15 hours from the PDMS removal

Differences in the ratios of 15 hours and 21 hours after the PDMS removal probably caused by images taken at different locations

Improvement suggestions

A microscope capable of capturing the entire well will allow better visualization of cell migration

Increasing the number of repetitions for each tested gelatin concentration would lead to more accurate and representative results

Investigating different seeding rates will allow an assessment of whether the initial seeding rate affects the resulted migration rates.

Questions?

References

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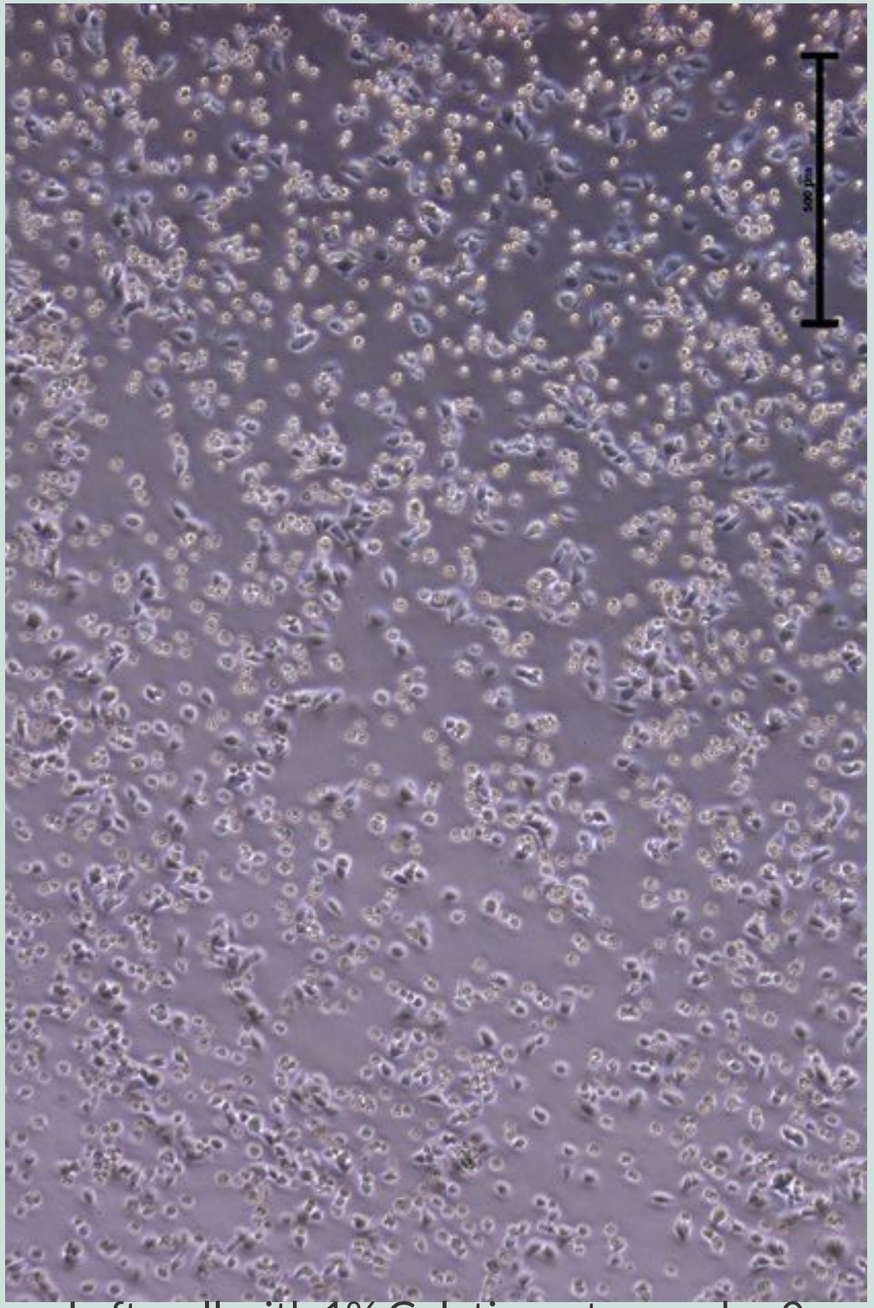
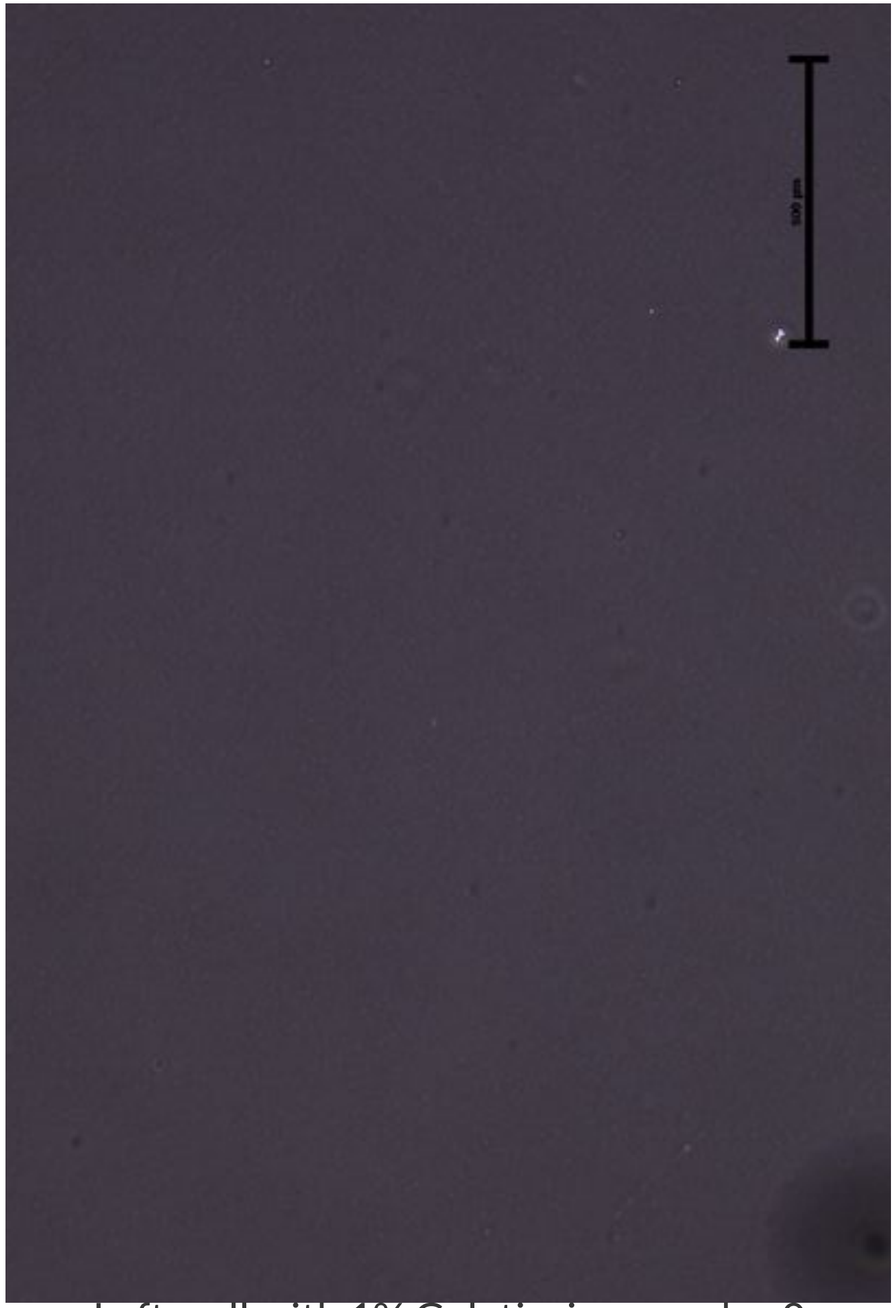
Outer Area



Left well with 1% Gelatin, out area, day 0

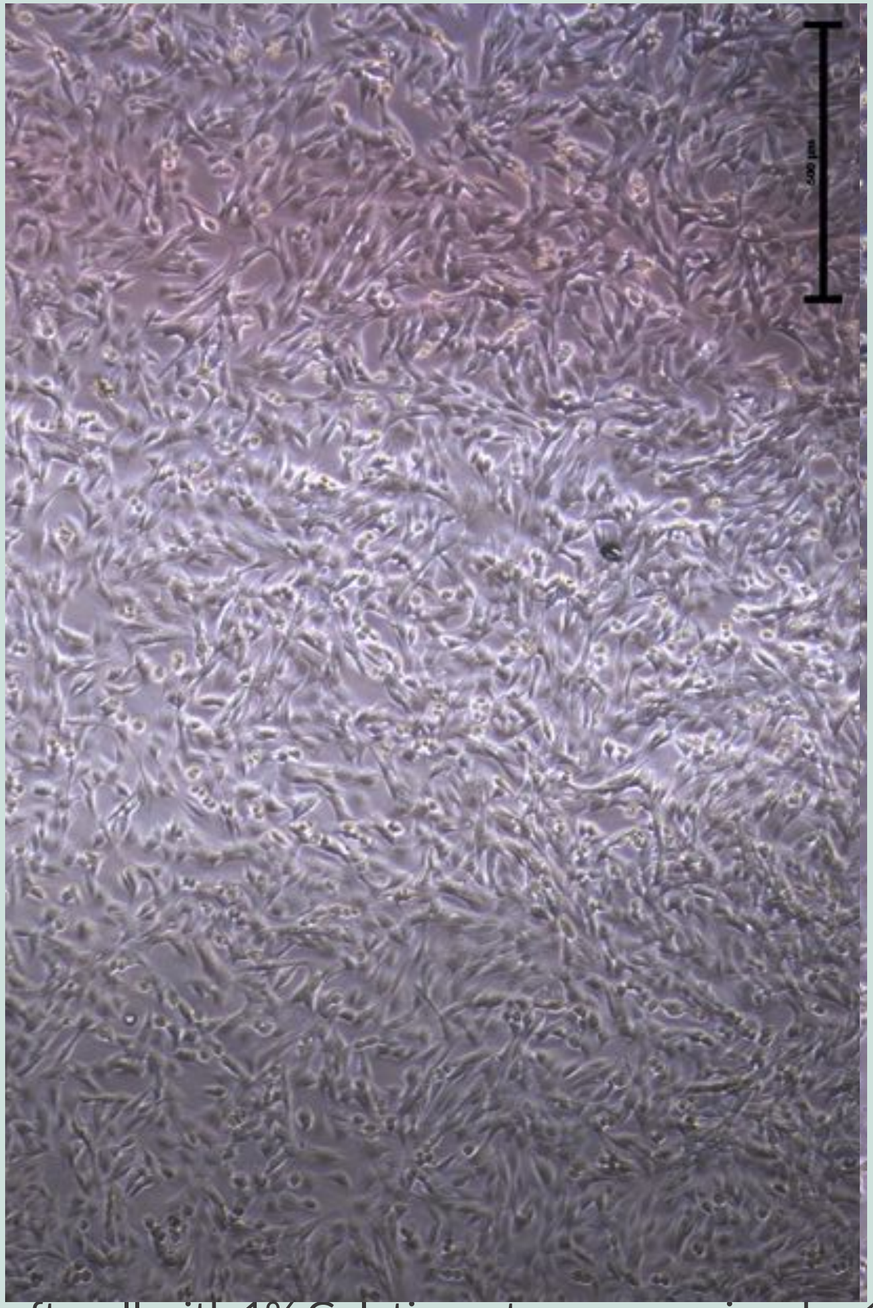
Inner Area



Left well with 1% Gelatin, in area, day 0

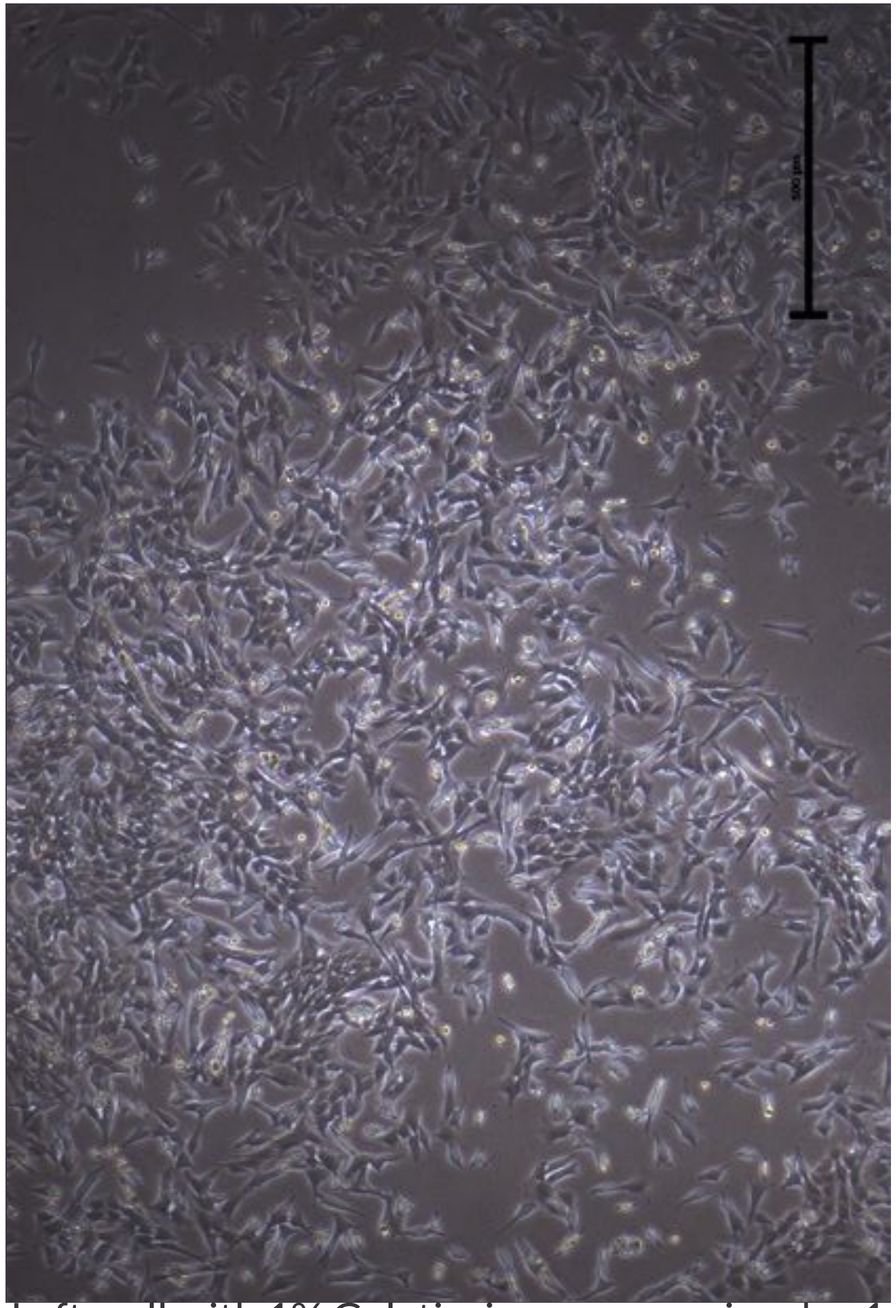
1% Gelatin Solution

Outer Area



Left well with 1% Gelatin, out area, morning day 1

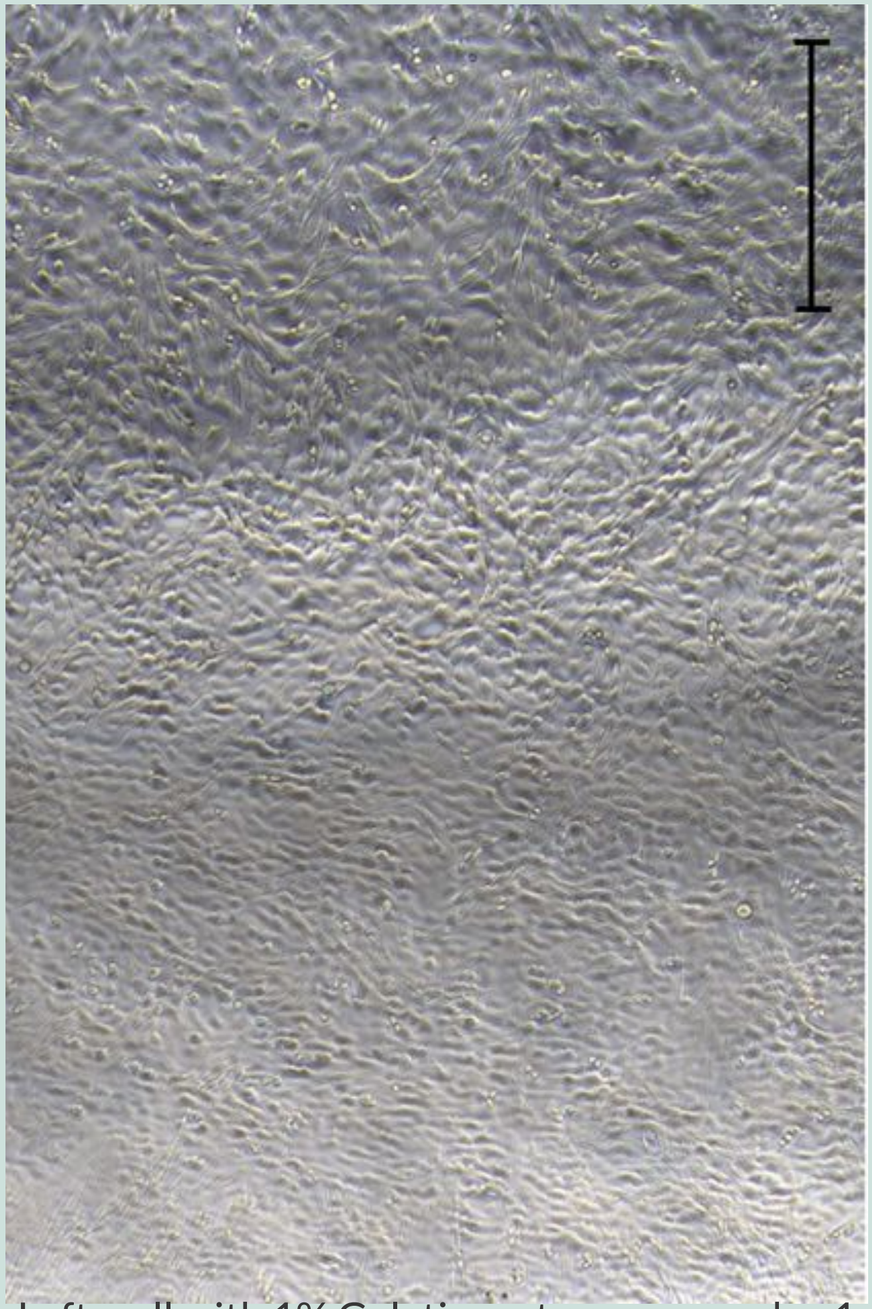
Inner Area



Left well with 1% Gelatin, in area, morning day 1

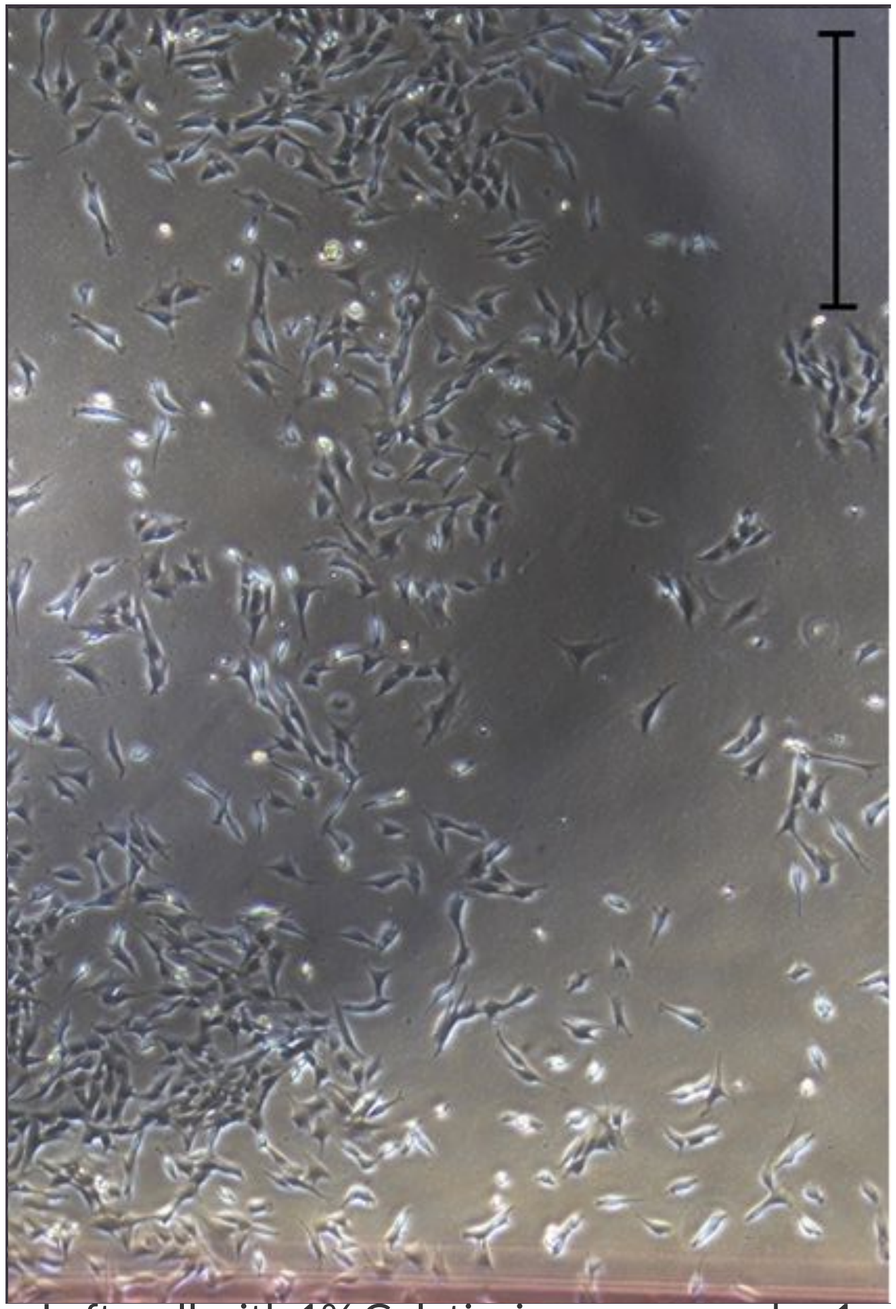
1% Gelatin Solution

Outer Area



Left well with 1% Gelatin, out area, noon day 1

Inner Area



Left well with 1% Gelatin, in area, noon day 1

1% Gelatin Solution